

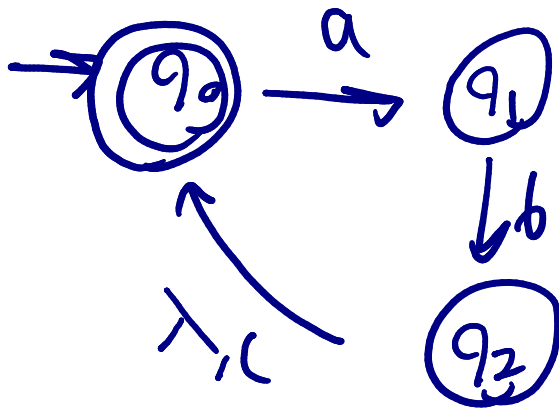
63011134 Chayut Lienlam

Theory of Computation

Submission link: <https://forms.gle/9QfcWiqewGmCcvG36>

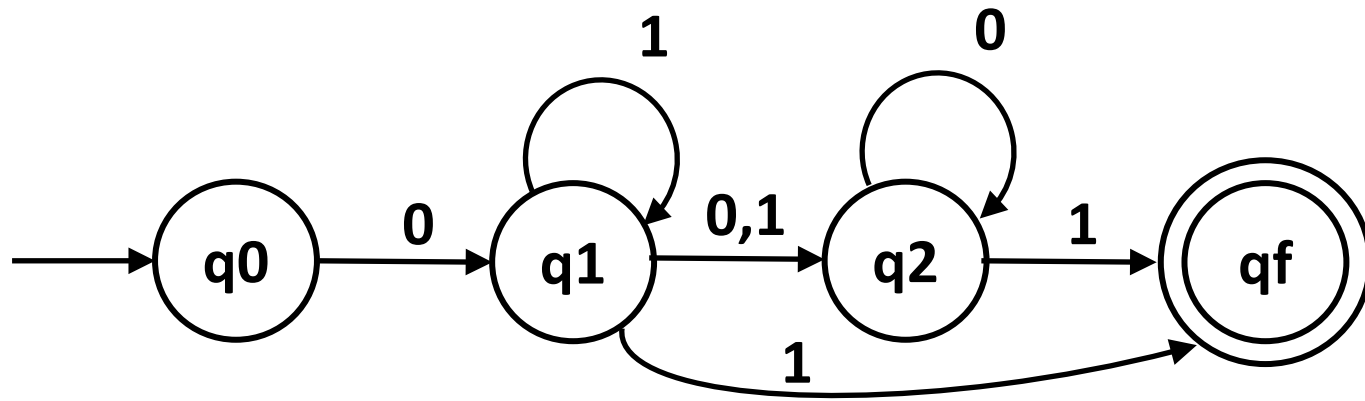
Exercise 3: (Nondeterministic Finite Automata - NFA)

1. Construct the minimal-state NFA that accepts the language $\{ab, abc\}^*$



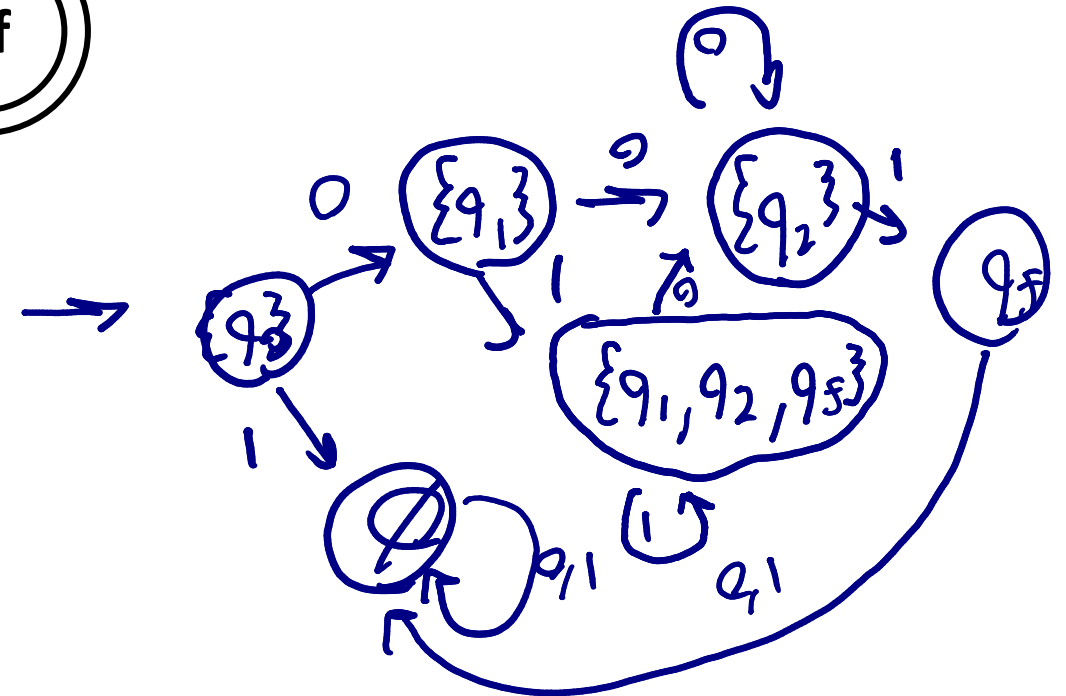
2. Convert the following NFA to DFA

NFA



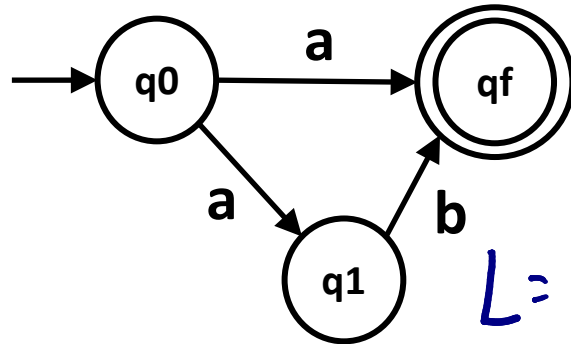
$01, 001, 011$

δ	0	1
q_0	$\{q_1\}$	$\emptyset$
q_1	$\{q_2\}$	$\{q_1, q_2, q_f\}$
q_2	$\{q_2\}$	$\{q_f\}$
q_f	$\emptyset$	$\emptyset$



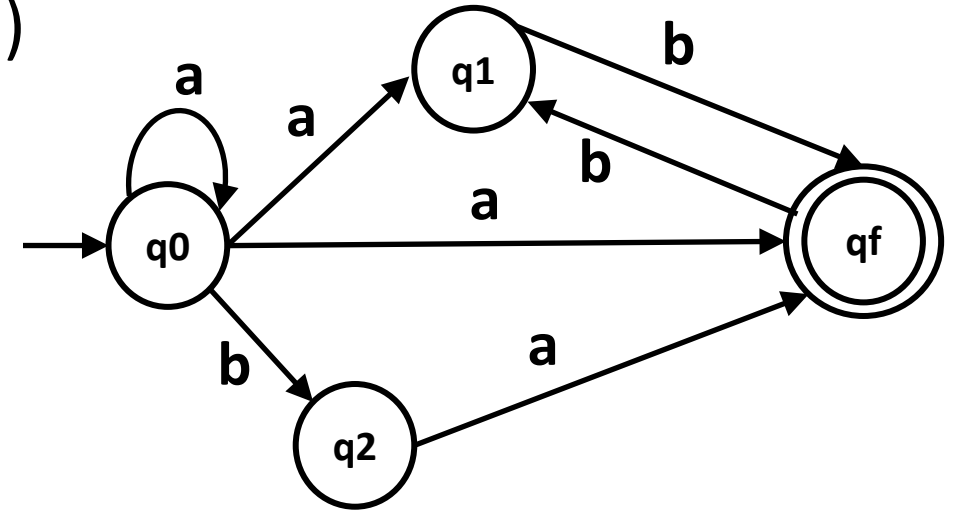
*3. What are the languages accepted by the following NFA ?

a)



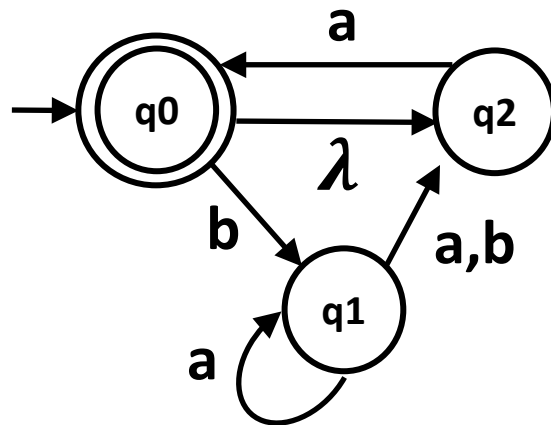
$L = \{a, ab\}$ ✓

b)



$\{a\}^* \bullet \{a, ba, ab\} \bullet \{bb\}^*$

c)



$L = \{a, ba^* \bullet \{a, b\} \bullet a\}^*$